

Detection of Epileptic Seizures through EEG Signals Using CNN :

- The primary objective of this project was to develop a model for the diagnosis of epilepsy using EEG signals to classify individuals into three distinct groups:
 1. Healthy
 2. Patients with seizures
 3. Patients without seizures
- The EEG data were obtained from the UKBonn Epileptologie Neurophysik Group: (<https://www.ukbonn.de/epileptologie/arbeitsgruppen/ag-lehnertz-neurophysik/downloads/>).
- A CNN model was employed in this project, including two convolutional layers with max-pooling, followed by fully connected layers.
- The model was trained using the Stochastic Gradient Descent (SGD) optimizer with a Cross-Entropy Loss criterion. The training process involves multiple epochs, with the model continuously improving its performance.
- After training, the model was evaluated on a separate test set, achieving an accuracy of 97.78%.

Please run the file “ EEG_CNN .ipynb “ in Colab.